

NCERT CLASS 11

60 Questions Per Topic

Based on the refined 2026–27 curriculum map created for this project.

Each curriculum topic below contains exactly 60 questions: understanding, examples/classification, comparison, application, reasoning/error analysis, and higher-order activity.

These are question-bank items derived from the topic descriptions in the supplied curriculum map. Validate against the exact school/textbook edition before high-stakes use.

MATHEMATICS — 1. Sets

Coverage: ■ Set types; subsets; operations; Venn diagrams; algebra of sets.

Understanding

1. What is ■ Set types, and what are its main features? Use a simple example.
2. What is subsets, and what are its main features? Use a school-life example.
3. What is operations, and what are its main features? Use a real-world example.
4. What is Venn diagrams, and what are its main features? Show the idea with a diagram where appropriate.
5. What is algebra of sets., and what are its main features? Explain in your own words.
6. What is ■ Set types, and what are its main features? Give a step-by-step response.
7. What is subsets, and what are its main features? Mention one common misconception.
8. What is operations, and what are its main features? State one practical application.
9. What is Venn diagrams, and what are its main features? Justify your response.
10. What is algebra of sets., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Set types and explain each. Use a simple example.
12. Give two examples related to subsets and explain each. Use a school-life example.
13. Give two examples related to operations and explain each. Use a real-world example.
14. Give two examples related to Venn diagrams and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to algebra of sets. and explain each. Explain in your own words.
16. Give two examples related to ■ Set types and explain each. Give a step-by-step response.
17. Give two examples related to subsets and explain each. Mention one common misconception.
18. Give two examples related to operations and explain each. State one practical application.
19. Give two examples related to Venn diagrams and explain each. Justify your response.
20. Give two examples related to algebra of sets. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Set types related to subsets? Compare the two ideas. Use a simple example.
22. How is subsets related to operations? Compare the two ideas. Use a school-life example.
23. How is operations related to Venn diagrams? Compare the two ideas. Use a real-world example.
24. How is Venn diagrams related to algebra of sets.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is algebra of sets. related to ■ Set types? Compare the two ideas. Explain in your own words.
26. How is ■ Set types related to subsets? Compare the two ideas. Give a step-by-step response.
27. How is subsets related to operations? Compare the two ideas. Mention one common misconception.
28. How is operations related to Venn diagrams? Compare the two ideas. State one practical application.
29. How is Venn diagrams related to algebra of sets.? Compare the two ideas. Justify your response.
30. How is algebra of sets. related to ■ Set types? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Set types to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply subsets to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply operations to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply Venn diagrams to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply algebra of sets. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Set types to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply subsets to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply operations to a realistic situation. What would you do or calculate? State one practical application.
39. Apply Venn diagrams to a realistic situation. What would you do or calculate? Justify your response.
40. Apply algebra of sets. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Set types. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of subsets. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of Venn diagrams. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of algebra of sets.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Set types. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of subsets. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of Venn diagrams. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of algebra of sets.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Set types. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show subsets. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show operations. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Venn diagrams. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show algebra of sets.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Set types. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show subsets. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show operations. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show Venn diagrams. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show algebra of sets.. Include the relevant rule or principle.

MATHEMATICS — 2. Relations and Functions

Coverage: ■ Cartesian product; relations; domain/range; functions; composition.

Understanding

1. What is ■ Cartesian product, and what are its main features? Use a simple example.
2. What is relations, and what are its main features? Use a school-life example.
3. What is domain/range, and what are its main features? Use a real-world example.
4. What is functions, and what are its main features? Show the idea with a diagram where appropriate.
5. What is composition., and what are its main features? Explain in your own words.
6. What is ■ Cartesian product, and what are its main features? Give a step-by-step response.
7. What is relations, and what are its main features? Mention one common misconception.
8. What is domain/range, and what are its main features? State one practical application.
9. What is functions, and what are its main features? Justify your response.
10. What is composition., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Cartesian product and explain each. Use a simple example.
12. Give two examples related to relations and explain each. Use a school-life example.
13. Give two examples related to domain/range and explain each. Use a real-world example.
14. Give two examples related to functions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to composition. and explain each. Explain in your own words.
16. Give two examples related to ■ Cartesian product and explain each. Give a step-by-step response.
17. Give two examples related to relations and explain each. Mention one common misconception.
18. Give two examples related to domain/range and explain each. State one practical application.
19. Give two examples related to functions and explain each. Justify your response.
20. Give two examples related to composition. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Cartesian product related to relations? Compare the two ideas. Use a simple example.
22. How is relations related to domain/range? Compare the two ideas. Use a school-life example.
23. How is domain/range related to functions? Compare the two ideas. Use a real-world example.
24. How is functions related to composition.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is composition. related to ■ Cartesian product? Compare the two ideas. Explain in your own words.
26. How is ■ Cartesian product related to relations? Compare the two ideas. Give a step-by-step response.
27. How is relations related to domain/range? Compare the two ideas. Mention one common misconception.
28. How is domain/range related to functions? Compare the two ideas. State one practical application.
29. How is functions related to composition.? Compare the two ideas. Justify your response.
30. How is composition. related to ■ Cartesian product? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Cartesian product to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply relations to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply domain/range to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply functions to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply composition. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Cartesian product to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply relations to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply domain/range to a realistic situation. What would you do or calculate? State one practical application.
39. Apply functions to a realistic situation. What would you do or calculate? Justify your response.
40. Apply composition. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Cartesian product. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of relations. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of domain/range. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of functions. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of composition.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Cartesian product. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of relations. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of domain/range. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of functions. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of composition.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cartesian product. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show relations. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show domain/range. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show functions. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show composition.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cartesian product. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show relations. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show domain/range. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show functions. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show composition.. Include the relevant rule or principle.

MATHEMATICS — 3. Trigonometric Functions

Coverage: ■ Ratios; identities; graphs; general solutions.

Understanding

1. What is ■ Ratios, and what are its main features? Use a simple example.
2. What is identities, and what are its main features? Use a school-life example.
3. What is graphs, and what are its main features? Use a real-world example.
4. What is general solutions., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Ratios, and what are its main features? Explain in your own words.
6. What is identities, and what are its main features? Give a step-by-step response.
7. What is graphs, and what are its main features? Mention one common misconception.
8. What is general solutions., and what are its main features? State one practical application.
9. What is ■ Ratios, and what are its main features? Justify your response.
10. What is identities, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to graphs and explain each. Use a simple example.
12. Give two examples related to general solutions. and explain each. Use a school-life example.
13. Give two examples related to ■ Ratios and explain each. Use a real-world example.
14. Give two examples related to identities and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to graphs and explain each. Explain in your own words.
16. Give two examples related to general solutions. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Ratios and explain each. Mention one common misconception.
18. Give two examples related to identities and explain each. State one practical application.
19. Give two examples related to graphs and explain each. Justify your response.
20. Give two examples related to general solutions. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Ratios related to identities? Compare the two ideas. Use a simple example.
22. How is identities related to graphs? Compare the two ideas. Use a school-life example.
23. How is graphs related to general solutions.? Compare the two ideas. Use a real-world example.
24. How is general solutions. related to ■ Ratios? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Ratios related to identities? Compare the two ideas. Explain in your own words.
26. How is identities related to graphs? Compare the two ideas. Give a step-by-step response.
27. How is graphs related to general solutions.? Compare the two ideas. Mention one common misconception.
28. How is general solutions. related to ■ Ratios? Compare the two ideas. State one practical application.
29. How is ■ Ratios related to identities? Compare the two ideas. Justify your response.
30. How is identities related to graphs? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply graphs to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply general solutions. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Ratios to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply identities to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply graphs to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply general solutions. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Ratios to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply identities to a realistic situation. What would you do or calculate? State one practical application.
39. Apply graphs to a realistic situation. What would you do or calculate? Justify your response.
40. Apply general solutions. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Ratios. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of graphs. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of general solutions.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Ratios. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of graphs. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of general solutions.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Ratios. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show general solutions.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ratios. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show identities. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show general solutions.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ratios. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show identities. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show general solutions.. Include the relevant rule or principle.

MATHEMATICS — 4. Complex Numbers and Quadratic Equations

Coverage: ■ Complex numbers; Argand plane; quadratic equations; roots.

Understanding

1. What is ■ Complex numbers, and what are its main features? Use a simple example.
2. What is Argand plane, and what are its main features? Use a school-life example.
3. What is quadratic equations, and what are its main features? Use a real-world example.
4. What is roots., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Complex numbers, and what are its main features? Explain in your own words.
6. What is Argand plane, and what are its main features? Give a step-by-step response.
7. What is quadratic equations, and what are its main features? Mention one common misconception.
8. What is roots., and what are its main features? State one practical application.
9. What is ■ Complex numbers, and what are its main features? Justify your response.
10. What is Argand plane, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to quadratic equations and explain each. Use a simple example.
12. Give two examples related to roots. and explain each. Use a school-life example.
13. Give two examples related to ■ Complex numbers and explain each. Use a real-world example.
14. Give two examples related to Argand plane and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to quadratic equations and explain each. Explain in your own words.
16. Give two examples related to roots. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Complex numbers and explain each. Mention one common misconception.
18. Give two examples related to Argand plane and explain each. State one practical application.
19. Give two examples related to quadratic equations and explain each. Justify your response.
20. Give two examples related to roots. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Complex numbers related to Argand plane? Compare the two ideas. Use a simple example.
22. How is Argand plane related to quadratic equations? Compare the two ideas. Use a school-life example.
23. How is quadratic equations related to roots.? Compare the two ideas. Use a real-world example.
24. How is roots. related to ■ Complex numbers? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Complex numbers related to Argand plane? Compare the two ideas. Explain in your own words.
26. How is Argand plane related to quadratic equations? Compare the two ideas. Give a step-by-step response.
27. How is quadratic equations related to roots.? Compare the two ideas. Mention one common misconception.
28. How is roots. related to ■ Complex numbers? Compare the two ideas. State one practical application.
29. How is ■ Complex numbers related to Argand plane? Compare the two ideas. Justify your response.
30. How is Argand plane related to quadratic equations? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply quadratic equations to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply roots. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Complex numbers to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply Argand plane to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply quadratic equations to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply roots. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Complex numbers to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply Argand plane to a realistic situation. What would you do or calculate? State one practical application.
39. Apply quadratic equations to a realistic situation. What would you do or calculate? Justify your response.
40. Apply roots. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Complex numbers. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of Argand plane. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of quadratic equations. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of roots.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Complex numbers. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of Argand plane. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of quadratic equations. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of roots.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Complex numbers. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of Argand plane. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show quadratic equations. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show roots.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Complex numbers. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Argand plane. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show quadratic equations. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show roots.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Complex numbers. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show Argand plane. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show quadratic equations. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show roots.. Include the relevant rule or principle.

MATHEMATICS — 5. Linear Inequalities

Coverage: ■ Algebraic and graphical solutions; interval representation.

Understanding

1. What is ■ Algebraic and graphical solutions, and what are its main features? Use a simple example.
2. What is interval representation., and what are its main features? Use a school-life example.
3. What is ■ Algebraic and graphical solutions, and what are its main features? Use a real-world example.
4. What is interval representation., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Algebraic and graphical solutions, and what are its main features? Explain in your own words.
6. What is interval representation., and what are its main features? Give a step-by-step response.
7. What is ■ Algebraic and graphical solutions, and what are its main features? Mention one common misconception.
8. What is interval representation., and what are its main features? State one practical application.
9. What is ■ Algebraic and graphical solutions, and what are its main features? Justify your response.
10. What is interval representation., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Algebraic and graphical solutions and explain each. Use a simple example.
12. Give two examples related to interval representation. and explain each. Use a school-life example.
13. Give two examples related to ■ Algebraic and graphical solutions and explain each. Use a real-world example.
14. Give two examples related to interval representation. and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to ■ Algebraic and graphical solutions and explain each. Explain in your own words.
16. Give two examples related to interval representation. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Algebraic and graphical solutions and explain each. Mention one common misconception.
18. Give two examples related to interval representation. and explain each. State one practical application.
19. Give two examples related to ■ Algebraic and graphical solutions and explain each. Justify your response.
20. Give two examples related to interval representation. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Algebraic and graphical solutions related to interval representation.? Compare the two ideas. Use a simple example.
22. How is interval representation. related to ■ Algebraic and graphical solutions? Compare the two ideas. Use a school-life example.
23. How is ■ Algebraic and graphical solutions related to interval representation.? Compare the two ideas. Use a real-world example.
24. How is interval representation. related to ■ Algebraic and graphical solutions? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Algebraic and graphical solutions related to interval representation.? Compare the two ideas. Explain in your own words.
26. How is interval representation. related to ■ Algebraic and graphical solutions? Compare the two ideas. Give a step-by-step response.
27. How is ■ Algebraic and graphical solutions related to interval representation.? Compare the two ideas. Mention one common misconception.
28. How is interval representation. related to ■ Algebraic and graphical solutions? Compare the two ideas. State one practical application.
29. How is ■ Algebraic and graphical solutions related to interval representation.? Compare the two ideas. Justify your response.
30. How is interval representation. related to ■ Algebraic and graphical solutions? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Algebraic and graphical solutions to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply interval representation. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Algebraic and graphical solutions to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply interval representation. to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply ■ Algebraic and graphical solutions to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply interval representation. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Algebraic and graphical solutions to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply interval representation. to a realistic situation. What would you do or calculate? State one practical application.
39. Apply ■ Algebraic and graphical solutions to a realistic situation. What would you do or calculate? Justify your response.
40. Apply interval representation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Algebraic and graphical solutions. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of interval representation.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Algebraic and graphical solutions. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of interval representation.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Algebraic and graphical solutions. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of interval representation.. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of ■ Algebraic and graphical solutions. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of interval representation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Algebraic and graphical solutions. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of interval representation.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Algebraic and graphical solutions. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show interval representation.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Algebraic and graphical solutions. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show interval representation.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Algebraic and graphical solutions. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show interval representation.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Algebraic and graphical solutions. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show interval representation.. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Algebraic and graphical solutions. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show interval representation.. Include the relevant rule or principle.

MATHEMATICS — 6. Permutations and Combinations

Coverage: ■ Counting principle; factorial; nPr ; nCr ; applications.

Understanding

1. What is ■ Counting principle, and what are its main features? Use a simple example.
2. What is factorial, and what are its main features? Use a school-life example.
3. What is nPr , and what are its main features? Use a real-world example.
4. What is nCr , and what are its main features? Show the idea with a diagram where appropriate.
5. What is applications., and what are its main features? Explain in your own words.
6. What is ■ Counting principle, and what are its main features? Give a step-by-step response.
7. What is factorial, and what are its main features? Mention one common misconception.
8. What is nPr , and what are its main features? State one practical application.
9. What is nCr , and what are its main features? Justify your response.
10. What is applications., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Counting principle and explain each. Use a simple example.
12. Give two examples related to factorial and explain each. Use a school-life example.
13. Give two examples related to nPr and explain each. Use a real-world example.
14. Give two examples related to nCr and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to applications. and explain each. Explain in your own words.
16. Give two examples related to ■ Counting principle and explain each. Give a step-by-step response.
17. Give two examples related to factorial and explain each. Mention one common misconception.
18. Give two examples related to nPr and explain each. State one practical application.
19. Give two examples related to nCr and explain each. Justify your response.
20. Give two examples related to applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Counting principle related to factorial? Compare the two ideas. Use a simple example.
22. How is factorial related to nPr ? Compare the two ideas. Use a school-life example.
23. How is nPr related to nCr ? Compare the two ideas. Use a real-world example.
24. How is nCr related to applications.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is applications. related to ■ Counting principle? Compare the two ideas. Explain in your own words.
26. How is ■ Counting principle related to factorial? Compare the two ideas. Give a step-by-step response.
27. How is factorial related to nPr ? Compare the two ideas. Mention one common misconception.
28. How is nPr related to nCr ? Compare the two ideas. State one practical application.
29. How is nCr related to applications.? Compare the two ideas. Justify your response.
30. How is applications. related to ■ Counting principle? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Counting principle to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply factorial to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply nPr to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply nCr to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply applications. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Counting principle to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply factorial to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply nPr to a realistic situation. What would you do or calculate? State one practical application.
39. Apply nCr to a realistic situation. What would you do or calculate? Justify your response.
40. Apply applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Counting principle. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of factorial. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of nPr . Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of nCr . Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Counting principle. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of factorial. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of nPr . Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of nCr . Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Counting principle. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show factorial. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show nPr . Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show nCr . Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Counting principle. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show factorial. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show nPr . State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show nCr . Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

MATHEMATICS — 7. Binomial Theorem

Coverage: ■ Expansion; general/middle terms; coefficients.

Understanding

1. What is ■ Expansion, and what are its main features? Use a simple example.
2. What is general/middle terms, and what are its main features? Use a school-life example.
3. What is coefficients., and what are its main features? Use a real-world example.
4. What is ■ Expansion, and what are its main features? Show the idea with a diagram where appropriate.
5. What is general/middle terms, and what are its main features? Explain in your own words.
6. What is coefficients., and what are its main features? Give a step-by-step response.
7. What is ■ Expansion, and what are its main features? Mention one common misconception.
8. What is general/middle terms, and what are its main features? State one practical application.
9. What is coefficients., and what are its main features? Justify your response.
10. What is ■ Expansion, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to general/middle terms and explain each. Use a simple example.
12. Give two examples related to coefficients. and explain each. Use a school-life example.
13. Give two examples related to ■ Expansion and explain each. Use a real-world example.
14. Give two examples related to general/middle terms and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to coefficients. and explain each. Explain in your own words.
16. Give two examples related to ■ Expansion and explain each. Give a step-by-step response.
17. Give two examples related to general/middle terms and explain each. Mention one common misconception.
18. Give two examples related to coefficients. and explain each. State one practical application.
19. Give two examples related to ■ Expansion and explain each. Justify your response.
20. Give two examples related to general/middle terms and explain each. Include the relevant rule or principle.

Comparison

21. How is coefficients. related to ■ Expansion? Compare the two ideas. Use a simple example.
22. How is ■ Expansion related to general/middle terms? Compare the two ideas. Use a school-life example.
23. How is general/middle terms related to coefficients.? Compare the two ideas. Use a real-world example.
24. How is coefficients. related to ■ Expansion? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Expansion related to general/middle terms? Compare the two ideas. Explain in your own words.
26. How is general/middle terms related to coefficients.? Compare the two ideas. Give a step-by-step response.
27. How is coefficients. related to ■ Expansion? Compare the two ideas. Mention one common misconception.
28. How is ■ Expansion related to general/middle terms? Compare the two ideas. State one practical application.
29. How is general/middle terms related to coefficients.? Compare the two ideas. Justify your response.
30. How is coefficients. related to ■ Expansion? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Expansion to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply general/middle terms to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply coefficients. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Expansion to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply general/middle terms to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply coefficients. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Expansion to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply general/middle terms to a realistic situation. What would you do or calculate? State one practical application.
39. Apply coefficients. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Expansion to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of general/middle terms. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of coefficients.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Expansion. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of general/middle terms. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of coefficients.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Expansion. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of general/middle terms. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of coefficients.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Expansion. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of general/middle terms. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show coefficients.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Expansion. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show general/middle terms. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show coefficients.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Expansion. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show general/middle terms. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show coefficients.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Expansion. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show general/middle terms. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show coefficients.. Include the relevant rule or principle.

MATHEMATICS — 8. Sequences and Series

Coverage: ■ AP/GP; sums; special series; applications.

Understanding

1. What is ■ AP/GP, and what are its main features? Use a simple example.
2. What is sums, and what are its main features? Use a school-life example.
3. What is special series, and what are its main features? Use a real-world example.
4. What is applications., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ AP/GP, and what are its main features? Explain in your own words.
6. What is sums, and what are its main features? Give a step-by-step response.
7. What is special series, and what are its main features? Mention one common misconception.
8. What is applications., and what are its main features? State one practical application.
9. What is ■ AP/GP, and what are its main features? Justify your response.
10. What is sums, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to special series and explain each. Use a simple example.
12. Give two examples related to applications. and explain each. Use a school-life example.
13. Give two examples related to ■ AP/GP and explain each. Use a real-world example.
14. Give two examples related to sums and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to special series and explain each. Explain in your own words.
16. Give two examples related to applications. and explain each. Give a step-by-step response.
17. Give two examples related to ■ AP/GP and explain each. Mention one common misconception.
18. Give two examples related to sums and explain each. State one practical application.
19. Give two examples related to special series and explain each. Justify your response.
20. Give two examples related to applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ AP/GP related to sums? Compare the two ideas. Use a simple example.
22. How is sums related to special series? Compare the two ideas. Use a school-life example.
23. How is special series related to applications.? Compare the two ideas. Use a real-world example.
24. How is applications. related to ■ AP/GP? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ AP/GP related to sums? Compare the two ideas. Explain in your own words.
26. How is sums related to special series? Compare the two ideas. Give a step-by-step response.
27. How is special series related to applications.? Compare the two ideas. Mention one common misconception.
28. How is applications. related to ■ AP/GP? Compare the two ideas. State one practical application.
29. How is ■ AP/GP related to sums? Compare the two ideas. Justify your response.
30. How is sums related to special series? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply special series to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply applications. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ AP/GP to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply sums to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply special series to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ AP/GP to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply sums to a realistic situation. What would you do or calculate? State one practical application.
39. Apply special series to a realistic situation. What would you do or calculate? Justify your response.
40. Apply applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ AP/GP. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of sums. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of special series. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ AP/GP. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of sums. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of special series. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ AP/GP. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of sums. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show special series. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ AP/GP. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show sums. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show special series. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ AP/GP. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show sums. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show special series. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

MATHEMATICS — 9. Straight Lines

Coverage: ■ Slope; line equations; angle between lines; distance from point.

Understanding

1. What is ■ Slope, and what are its main features? Use a simple example.
2. What is line equations, and what are its main features? Use a school-life example.
3. What is angle between lines, and what are its main features? Use a real-world example.
4. What is distance from point., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Slope, and what are its main features? Explain in your own words.
6. What is line equations, and what are its main features? Give a step-by-step response.
7. What is angle between lines, and what are its main features? Mention one common misconception.
8. What is distance from point., and what are its main features? State one practical application.
9. What is ■ Slope, and what are its main features? Justify your response.
10. What is line equations, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to angle between lines and explain each. Use a simple example.
12. Give two examples related to distance from point. and explain each. Use a school-life example.
13. Give two examples related to ■ Slope and explain each. Use a real-world example.
14. Give two examples related to line equations and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to angle between lines and explain each. Explain in your own words.
16. Give two examples related to distance from point. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Slope and explain each. Mention one common misconception.
18. Give two examples related to line equations and explain each. State one practical application.
19. Give two examples related to angle between lines and explain each. Justify your response.
20. Give two examples related to distance from point. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Slope related to line equations? Compare the two ideas. Use a simple example.
22. How is line equations related to angle between lines? Compare the two ideas. Use a school-life example.
23. How is angle between lines related to distance from point.? Compare the two ideas. Use a real-world example.
24. How is distance from point. related to ■ Slope? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Slope related to line equations? Compare the two ideas. Explain in your own words.
26. How is line equations related to angle between lines? Compare the two ideas. Give a step-by-step response.
27. How is angle between lines related to distance from point.? Compare the two ideas. Mention one common misconception.
28. How is distance from point. related to ■ Slope? Compare the two ideas. State one practical application.
29. How is ■ Slope related to line equations? Compare the two ideas. Justify your response.
30. How is line equations related to angle between lines? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply angle between lines to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply distance from point. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Slope to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply line equations to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply angle between lines to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply distance from point. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Slope to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply line equations to a realistic situation. What would you do or calculate? State one practical application.
39. Apply angle between lines to a realistic situation. What would you do or calculate? Justify your response.
40. Apply distance from point. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Slope. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of line equations. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of angle between lines. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of distance from point.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Slope. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of line equations. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of angle between lines. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of distance from point.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Slope. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of line equations. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show angle between lines. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show distance from point.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Slope. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show line equations. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show angle between lines. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show distance from point.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Slope. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show line equations. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show angle between lines. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show distance from point.. Include the relevant rule or principle.

MATHEMATICS — 10. Conic Sections

Coverage: ■ Circle; parabola; ellipse; hyperbola; standard equations.

Understanding

1. What is ■ Circle, and what are its main features? Use a simple example.
2. What is parabola, and what are its main features? Use a school-life example.
3. What is ellipse, and what are its main features? Use a real-world example.
4. What is hyperbola, and what are its main features? Show the idea with a diagram where appropriate.
5. What is standard equations., and what are its main features? Explain in your own words.
6. What is ■ Circle, and what are its main features? Give a step-by-step response.
7. What is parabola, and what are its main features? Mention one common misconception.
8. What is ellipse, and what are its main features? State one practical application.
9. What is hyperbola, and what are its main features? Justify your response.
10. What is standard equations., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Circle and explain each. Use a simple example.
12. Give two examples related to parabola and explain each. Use a school-life example.
13. Give two examples related to ellipse and explain each. Use a real-world example.
14. Give two examples related to hyperbola and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to standard equations. and explain each. Explain in your own words.
16. Give two examples related to ■ Circle and explain each. Give a step-by-step response.
17. Give two examples related to parabola and explain each. Mention one common misconception.
18. Give two examples related to ellipse and explain each. State one practical application.
19. Give two examples related to hyperbola and explain each. Justify your response.
20. Give two examples related to standard equations. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Circle related to parabola? Compare the two ideas. Use a simple example.
22. How is parabola related to ellipse? Compare the two ideas. Use a school-life example.
23. How is ellipse related to hyperbola? Compare the two ideas. Use a real-world example.
24. How is hyperbola related to standard equations.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is standard equations. related to ■ Circle? Compare the two ideas. Explain in your own words.
26. How is ■ Circle related to parabola? Compare the two ideas. Give a step-by-step response.
27. How is parabola related to ellipse? Compare the two ideas. Mention one common misconception.
28. How is ellipse related to hyperbola? Compare the two ideas. State one practical application.
29. How is hyperbola related to standard equations.? Compare the two ideas. Justify your response.
30. How is standard equations. related to ■ Circle? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Circle to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply parabola to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ellipse to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply hyperbola to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply standard equations. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Circle to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply parabola to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply ellipse to a realistic situation. What would you do or calculate? State one practical application.
39. Apply hyperbola to a realistic situation. What would you do or calculate? Justify your response.
40. Apply standard equations. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of $\blacksquare$ Circle. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of parabola. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ellipse. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of hyperbola. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of standard equations.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of $\blacksquare$ Circle. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of parabola. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of ellipse. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of hyperbola. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of standard equations.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show $\blacksquare$ Circle. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show parabola. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ellipse. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show hyperbola. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show standard equations.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show $\blacksquare$ Circle. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show parabola. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ellipse. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show hyperbola. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show standard equations.. Include the relevant rule or principle.

MATHEMATICS — 11. Introduction to Three-Dimensional Geometry

Coverage: ■ Coordinates; distance; geometric interpretation.

Understanding

1. What is ■ Coordinates, and what are its main features? Use a simple example.
2. What is distance, and what are its main features? Use a school-life example.
3. What is geometric interpretation., and what are its main features? Use a real-world example.
4. What is ■ Coordinates, and what are its main features? Show the idea with a diagram where appropriate.
5. What is distance, and what are its main features? Explain in your own words.
6. What is geometric interpretation., and what are its main features? Give a step-by-step response.
7. What is ■ Coordinates, and what are its main features? Mention one common misconception.
8. What is distance, and what are its main features? State one practical application.
9. What is geometric interpretation., and what are its main features? Justify your response.
10. What is ■ Coordinates, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to distance and explain each. Use a simple example.
12. Give two examples related to geometric interpretation. and explain each. Use a school-life example.
13. Give two examples related to ■ Coordinates and explain each. Use a real-world example.
14. Give two examples related to distance and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to geometric interpretation. and explain each. Explain in your own words.
16. Give two examples related to ■ Coordinates and explain each. Give a step-by-step response.
17. Give two examples related to distance and explain each. Mention one common misconception.
18. Give two examples related to geometric interpretation. and explain each. State one practical application.
19. Give two examples related to ■ Coordinates and explain each. Justify your response.
20. Give two examples related to distance and explain each. Include the relevant rule or principle.

Comparison

21. How is geometric interpretation. related to ■ Coordinates? Compare the two ideas. Use a simple example.
22. How is ■ Coordinates related to distance? Compare the two ideas. Use a school-life example.
23. How is distance related to geometric interpretation.? Compare the two ideas. Use a real-world example.
24. How is geometric interpretation. related to ■ Coordinates? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Coordinates related to distance? Compare the two ideas. Explain in your own words.
26. How is distance related to geometric interpretation.? Compare the two ideas. Give a step-by-step response.
27. How is geometric interpretation. related to ■ Coordinates? Compare the two ideas. Mention one common misconception.
28. How is ■ Coordinates related to distance? Compare the two ideas. State one practical application.
29. How is distance related to geometric interpretation.? Compare the two ideas. Justify your response.
30. How is geometric interpretation. related to ■ Coordinates? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Coordinates to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply distance to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply geometric interpretation. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Coordinates to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply distance to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply geometric interpretation. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Coordinates to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply distance to a realistic situation. What would you do or calculate? State one practical application.
39. Apply geometric interpretation. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Coordinates to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of geometric interpretation.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Coordinates. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of geometric interpretation.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Coordinates. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of geometric interpretation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Coordinates. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show geometric interpretation.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Coordinates. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show geometric interpretation.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Coordinates. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show geometric interpretation.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Coordinates. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show geometric interpretation.. Include the relevant rule or principle.

MATHEMATICS — 12. Limits and Derivatives

Coverage: ■ Limits; continuity intuition; derivatives; basic rules.

Understanding

1. What is ■ Limits, and what are its main features? Use a simple example.
2. What is continuity intuition, and what are its main features? Use a school-life example.
3. What is derivatives, and what are its main features? Use a real-world example.
4. What is basic rules., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Limits, and what are its main features? Explain in your own words.
6. What is continuity intuition, and what are its main features? Give a step-by-step response.
7. What is derivatives, and what are its main features? Mention one common misconception.
8. What is basic rules., and what are its main features? State one practical application.
9. What is ■ Limits, and what are its main features? Justify your response.
10. What is continuity intuition, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to derivatives and explain each. Use a simple example.
12. Give two examples related to basic rules. and explain each. Use a school-life example.
13. Give two examples related to ■ Limits and explain each. Use a real-world example.
14. Give two examples related to continuity intuition and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to derivatives and explain each. Explain in your own words.
16. Give two examples related to basic rules. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Limits and explain each. Mention one common misconception.
18. Give two examples related to continuity intuition and explain each. State one practical application.
19. Give two examples related to derivatives and explain each. Justify your response.
20. Give two examples related to basic rules. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Limits related to continuity intuition? Compare the two ideas. Use a simple example.
22. How is continuity intuition related to derivatives? Compare the two ideas. Use a school-life example.
23. How is derivatives related to basic rules.? Compare the two ideas. Use a real-world example.
24. How is basic rules. related to ■ Limits? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Limits related to continuity intuition? Compare the two ideas. Explain in your own words.
26. How is continuity intuition related to derivatives? Compare the two ideas. Give a step-by-step response.
27. How is derivatives related to basic rules.? Compare the two ideas. Mention one common misconception.
28. How is basic rules. related to ■ Limits? Compare the two ideas. State one practical application.
29. How is ■ Limits related to continuity intuition? Compare the two ideas. Justify your response.
30. How is continuity intuition related to derivatives? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply derivatives to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply basic rules. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Limits to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply continuity intuition to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply derivatives to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply basic rules. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Limits to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply continuity intuition to a realistic situation. What would you do or calculate? State one practical application.
39. Apply derivatives to a realistic situation. What would you do or calculate? Justify your response.
40. Apply basic rules. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Limits. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of continuity intuition. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of derivatives. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of basic rules.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Limits. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of continuity intuition. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of derivatives. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of basic rules.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Limits. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of continuity intuition. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show derivatives. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show basic rules.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Limits. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show continuity intuition. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show derivatives. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show basic rules.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Limits. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show continuity intuition. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show derivatives. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show basic rules.. Include the relevant rule or principle.

MATHEMATICS — 13. Statistics

Coverage: ■ Dispersion; variance; standard deviation; interpretation.

Understanding

1. What is ■ Dispersion, and what are its main features? Use a simple example.
2. What is variance, and what are its main features? Use a school-life example.
3. What is standard deviation, and what are its main features? Use a real-world example.
4. What is interpretation., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Dispersion, and what are its main features? Explain in your own words.
6. What is variance, and what are its main features? Give a step-by-step response.
7. What is standard deviation, and what are its main features? Mention one common misconception.
8. What is interpretation., and what are its main features? State one practical application.
9. What is ■ Dispersion, and what are its main features? Justify your response.
10. What is variance, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to standard deviation and explain each. Use a simple example.
12. Give two examples related to interpretation. and explain each. Use a school-life example.
13. Give two examples related to ■ Dispersion and explain each. Use a real-world example.
14. Give two examples related to variance and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to standard deviation and explain each. Explain in your own words.
16. Give two examples related to interpretation. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Dispersion and explain each. Mention one common misconception.
18. Give two examples related to variance and explain each. State one practical application.
19. Give two examples related to standard deviation and explain each. Justify your response.
20. Give two examples related to interpretation. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Dispersion related to variance? Compare the two ideas. Use a simple example.
22. How is variance related to standard deviation? Compare the two ideas. Use a school-life example.
23. How is standard deviation related to interpretation.? Compare the two ideas. Use a real-world example.
24. How is interpretation. related to ■ Dispersion? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Dispersion related to variance? Compare the two ideas. Explain in your own words.
26. How is variance related to standard deviation? Compare the two ideas. Give a step-by-step response.
27. How is standard deviation related to interpretation.? Compare the two ideas. Mention one common misconception.
28. How is interpretation. related to ■ Dispersion? Compare the two ideas. State one practical application.
29. How is ■ Dispersion related to variance? Compare the two ideas. Justify your response.
30. How is variance related to standard deviation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply standard deviation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply interpretation. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Dispersion to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply variance to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply standard deviation to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply interpretation. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Dispersion to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply variance to a realistic situation. What would you do or calculate? State one practical application.
39. Apply standard deviation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply interpretation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Dispersion. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of variance. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of standard deviation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of interpretation.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Dispersion. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of variance. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of standard deviation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of interpretation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Dispersion. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of variance. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show standard deviation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Dispersion. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show variance. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show standard deviation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Dispersion. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show variance. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show standard deviation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation.. Include the relevant rule or principle.

MATHEMATICS — 14. Probability

Coverage: ■ Random experiments; events; axiomatic probability; applications.

Understanding

1. What is ■ Random experiments, and what are its main features? Use a simple example.
2. What is events, and what are its main features? Use a school-life example.
3. What is axiomatic probability, and what are its main features? Use a real-world example.
4. What is applications., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Random experiments, and what are its main features? Explain in your own words.
6. What is events, and what are its main features? Give a step-by-step response.
7. What is axiomatic probability, and what are its main features? Mention one common misconception.
8. What is applications., and what are its main features? State one practical application.
9. What is ■ Random experiments, and what are its main features? Justify your response.
10. What is events, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to axiomatic probability and explain each. Use a simple example.
12. Give two examples related to applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Random experiments and explain each. Use a real-world example.
14. Give two examples related to events and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to axiomatic probability and explain each. Explain in your own words.
16. Give two examples related to applications. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Random experiments and explain each. Mention one common misconception.
18. Give two examples related to events and explain each. State one practical application.
19. Give two examples related to axiomatic probability and explain each. Justify your response.
20. Give two examples related to applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Random experiments related to events? Compare the two ideas. Use a simple example.
22. How is events related to axiomatic probability? Compare the two ideas. Use a school-life example.
23. How is axiomatic probability related to applications.? Compare the two ideas. Use a real-world example.
24. How is applications. related to ■ Random experiments? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Random experiments related to events? Compare the two ideas. Explain in your own words.
26. How is events related to axiomatic probability? Compare the two ideas. Give a step-by-step response.
27. How is axiomatic probability related to applications.? Compare the two ideas. Mention one common misconception.
28. How is applications. related to ■ Random experiments? Compare the two ideas. State one practical application.
29. How is ■ Random experiments related to events? Compare the two ideas. Justify your response.
30. How is events related to axiomatic probability? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply axiomatic probability to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply applications. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Random experiments to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply events to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply axiomatic probability to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Random experiments to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply events to a realistic situation. What would you do or calculate? State one practical application.
39. Apply axiomatic probability to a realistic situation. What would you do or calculate? Justify your response.
40. Apply applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Random experiments. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of events. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of axiomatic probability. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Random experiments. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of events. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of axiomatic probability. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Random experiments. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of events. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show axiomatic probability. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Random experiments. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show events. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show axiomatic probability. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Random experiments. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show events. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show axiomatic probability. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 1. Units and Measurements

Coverage: ■ SI units; dimensions; significant figures; errors.

Understanding

1. What is ■ SI units, and what are its main features? Use a simple example.
2. What is dimensions, and what are its main features? Use a school-life example.
3. What is significant figures, and what are its main features? Use a real-world example.
4. What is errors., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ SI units, and what are its main features? Explain in your own words.
6. What is dimensions, and what are its main features? Give a step-by-step response.
7. What is significant figures, and what are its main features? Mention one common misconception.
8. What is errors., and what are its main features? State one practical application.
9. What is ■ SI units, and what are its main features? Justify your response.
10. What is dimensions, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to significant figures and explain each. Use a simple example.
12. Give two examples related to errors. and explain each. Use a school-life example.
13. Give two examples related to ■ SI units and explain each. Use a real-world example.
14. Give two examples related to dimensions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to significant figures and explain each. Explain in your own words.
16. Give two examples related to errors. and explain each. Give a step-by-step response.
17. Give two examples related to ■ SI units and explain each. Mention one common misconception.
18. Give two examples related to dimensions and explain each. State one practical application.
19. Give two examples related to significant figures and explain each. Justify your response.
20. Give two examples related to errors. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ SI units related to dimensions? Compare the two ideas. Use a simple example.
22. How is dimensions related to significant figures? Compare the two ideas. Use a school-life example.
23. How is significant figures related to errors.? Compare the two ideas. Use a real-world example.
24. How is errors. related to ■ SI units? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ SI units related to dimensions? Compare the two ideas. Explain in your own words.
26. How is dimensions related to significant figures? Compare the two ideas. Give a step-by-step response.
27. How is significant figures related to errors.? Compare the two ideas. Mention one common misconception.
28. How is errors. related to ■ SI units? Compare the two ideas. State one practical application.
29. How is ■ SI units related to dimensions? Compare the two ideas. Justify your response.
30. How is dimensions related to significant figures? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply significant figures to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply errors. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ SI units to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply dimensions to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply significant figures to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply errors. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ SI units to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply dimensions to a realistic situation. What would you do or calculate? State one practical application.
39. Apply significant figures to a realistic situation. What would you do or calculate? Justify your response.
40. Apply errors. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ SI units. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of dimensions. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of significant figures. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of errors.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ SI units. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of dimensions. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of significant figures. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of errors.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ SI units. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of dimensions. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show significant figures. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show errors.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ SI units. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show dimensions. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show significant figures. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show errors.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ SI units. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show dimensions. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show significant figures. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show errors.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 2. Motion in a Straight Line

Coverage: ■ Position; velocity; acceleration; graphs; equations.

Understanding

1. What is ■ Position, and what are its main features? Use a simple example.
2. What is velocity, and what are its main features? Use a school-life example.
3. What is acceleration, and what are its main features? Use a real-world example.
4. What is graphs, and what are its main features? Show the idea with a diagram where appropriate.
5. What is equations., and what are its main features? Explain in your own words.
6. What is ■ Position, and what are its main features? Give a step-by-step response.
7. What is velocity, and what are its main features? Mention one common misconception.
8. What is acceleration, and what are its main features? State one practical application.
9. What is graphs, and what are its main features? Justify your response.
10. What is equations., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Position and explain each. Use a simple example.
12. Give two examples related to velocity and explain each. Use a school-life example.
13. Give two examples related to acceleration and explain each. Use a real-world example.
14. Give two examples related to graphs and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to equations. and explain each. Explain in your own words.
16. Give two examples related to ■ Position and explain each. Give a step-by-step response.
17. Give two examples related to velocity and explain each. Mention one common misconception.
18. Give two examples related to acceleration and explain each. State one practical application.
19. Give two examples related to graphs and explain each. Justify your response.
20. Give two examples related to equations. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Position related to velocity? Compare the two ideas. Use a simple example.
22. How is velocity related to acceleration? Compare the two ideas. Use a school-life example.
23. How is acceleration related to graphs? Compare the two ideas. Use a real-world example.
24. How is graphs related to equations.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is equations. related to ■ Position? Compare the two ideas. Explain in your own words.
26. How is ■ Position related to velocity? Compare the two ideas. Give a step-by-step response.
27. How is velocity related to acceleration? Compare the two ideas. Mention one common misconception.
28. How is acceleration related to graphs? Compare the two ideas. State one practical application.
29. How is graphs related to equations.? Compare the two ideas. Justify your response.
30. How is equations. related to ■ Position? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Position to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply velocity to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply acceleration to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply graphs to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply equations. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Position to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply velocity to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply acceleration to a realistic situation. What would you do or calculate? State one practical application.
39. Apply graphs to a realistic situation. What would you do or calculate? Justify your response.
40. Apply equations. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Position. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of velocity. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of acceleration. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of graphs. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of equations.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Position. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of velocity. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of acceleration. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of graphs. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of equations.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Position. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show velocity. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show acceleration. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show equations.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Position. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show velocity. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show acceleration. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show equations.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 3. Motion in a Plane

Coverage: ■ Vectors; projectile motion; circular motion.

Understanding

1. What is ■ Vectors, and what are its main features? Use a simple example.
2. What is projectile motion, and what are its main features? Use a school-life example.
3. What is circular motion., and what are its main features? Use a real-world example.
4. What is ■ Vectors, and what are its main features? Show the idea with a diagram where appropriate.
5. What is projectile motion, and what are its main features? Explain in your own words.
6. What is circular motion., and what are its main features? Give a step-by-step response.
7. What is ■ Vectors, and what are its main features? Mention one common misconception.
8. What is projectile motion, and what are its main features? State one practical application.
9. What is circular motion., and what are its main features? Justify your response.
10. What is ■ Vectors, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to projectile motion and explain each. Use a simple example.
12. Give two examples related to circular motion. and explain each. Use a school-life example.
13. Give two examples related to ■ Vectors and explain each. Use a real-world example.
14. Give two examples related to projectile motion and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to circular motion. and explain each. Explain in your own words.
16. Give two examples related to ■ Vectors and explain each. Give a step-by-step response.
17. Give two examples related to projectile motion and explain each. Mention one common misconception.
18. Give two examples related to circular motion. and explain each. State one practical application.
19. Give two examples related to ■ Vectors and explain each. Justify your response.
20. Give two examples related to projectile motion and explain each. Include the relevant rule or principle.

Comparison

21. How is circular motion. related to ■ Vectors? Compare the two ideas. Use a simple example.
22. How is ■ Vectors related to projectile motion? Compare the two ideas. Use a school-life example.
23. How is projectile motion related to circular motion.? Compare the two ideas. Use a real-world example.
24. How is circular motion. related to ■ Vectors? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Vectors related to projectile motion? Compare the two ideas. Explain in your own words.
26. How is projectile motion related to circular motion.? Compare the two ideas. Give a step-by-step response.
27. How is circular motion. related to ■ Vectors? Compare the two ideas. Mention one common misconception.
28. How is ■ Vectors related to projectile motion? Compare the two ideas. State one practical application.
29. How is projectile motion related to circular motion.? Compare the two ideas. Justify your response.
30. How is circular motion. related to ■ Vectors? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Vectors to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply projectile motion to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply circular motion. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Vectors to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply projectile motion to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply circular motion. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Vectors to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply projectile motion to a realistic situation. What would you do or calculate? State one practical application.
39. Apply circular motion. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Vectors to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of projectile motion. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of circular motion.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Vectors. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of projectile motion. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of circular motion.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Vectors. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of projectile motion. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of circular motion.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Vectors. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of projectile motion. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show circular motion.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Vectors. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show projectile motion. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show circular motion.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Vectors. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show projectile motion. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show circular motion.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Vectors. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show projectile motion. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show circular motion.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 4. Laws of Motion

Coverage: NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus

Understanding

1. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a simple example.
2. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a school-life example.
3. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a real-world example.
4. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Show the idea with a diagram where appropriate.
5. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Explain in your own words.
6. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Give a step-by-step response.
7. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Mention one common misconception.
8. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? State one practical application.
9. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Justify your response.
10. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a simple example.
12. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a school-life example.
13. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a real-world example.
14. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Explain in your own words.
16. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Give a step-by-step response.
17. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Mention one common misconception.
18. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. State one practical application.
19. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Justify your response.
20. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Include the relevant rule or principle.

Comparison

21. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a simple example.
22. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a school-life example.

- 23.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a real-world example.
- 24.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Show the idea with a diagram where appropriate.
- 25.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Explain in your own words.
- 26.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Give a step-by-step response.
- 27.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Mention one common misconception.
- 28.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. State one practical application.
- 29.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Justify your response.
- 30.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Include the relevant rule or principle.

Application

- 31.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a simple example.
- 32.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a school-life example.
- 33.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a real-world example.
- 34.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
- 35.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Explain in your own words.
- 36.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Give a step-by-step response.
- 37.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Mention one common misconception.
- 38.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? State one practical application.
- 39.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Justify your response.
- 40.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

- 41.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a simple example.
- 42.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a school-life example.
- 43.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a real-world example.
- 44.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
- 45.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Explain in your own words.
- 46.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Give a step-by-step response.
- 47.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 5. Work, Energy and Power

Coverage: ■ Work-energy theorem; potential/kinetic energy; conservation; power.

Understanding

1. What is ■ Work-energy theorem, and what are its main features? Use a simple example.
2. What is potential/kinetic energy, and what are its main features? Use a school-life example.
3. What is conservation, and what are its main features? Use a real-world example.
4. What is power., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Work-energy theorem, and what are its main features? Explain in your own words.
6. What is potential/kinetic energy, and what are its main features? Give a step-by-step response.
7. What is conservation, and what are its main features? Mention one common misconception.
8. What is power., and what are its main features? State one practical application.
9. What is ■ Work-energy theorem, and what are its main features? Justify your response.
10. What is potential/kinetic energy, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to conservation and explain each. Use a simple example.
12. Give two examples related to power. and explain each. Use a school-life example.
13. Give two examples related to ■ Work-energy theorem and explain each. Use a real-world example.
14. Give two examples related to potential/kinetic energy and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to conservation and explain each. Explain in your own words.
16. Give two examples related to power. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Work-energy theorem and explain each. Mention one common misconception.
18. Give two examples related to potential/kinetic energy and explain each. State one practical application.
19. Give two examples related to conservation and explain each. Justify your response.
20. Give two examples related to power. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Work-energy theorem related to potential/kinetic energy? Compare the two ideas. Use a simple example.
22. How is potential/kinetic energy related to conservation? Compare the two ideas. Use a school-life example.
23. How is conservation related to power.? Compare the two ideas. Use a real-world example.
24. How is power. related to ■ Work-energy theorem? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Work-energy theorem related to potential/kinetic energy? Compare the two ideas. Explain in your own words.
26. How is potential/kinetic energy related to conservation? Compare the two ideas. Give a step-by-step response.
27. How is conservation related to power.? Compare the two ideas. Mention one common misconception.
28. How is power. related to ■ Work-energy theorem? Compare the two ideas. State one practical application.
29. How is ■ Work-energy theorem related to potential/kinetic energy? Compare the two ideas. Justify your response.
30. How is potential/kinetic energy related to conservation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply conservation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply power. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Work-energy theorem to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply potential/kinetic energy to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply conservation to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply power. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Work-energy theorem to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply potential/kinetic energy to a realistic situation. What would you do or calculate? State one practical application.
39. Apply conservation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply power. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Work-energy theorem. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of potential/kinetic energy. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of conservation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of power.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Work-energy theorem. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of potential/kinetic energy. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of conservation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of power.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Work-energy theorem. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of potential/kinetic energy. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show conservation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show power.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Work-energy theorem. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show potential/kinetic energy. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show conservation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show power.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Work-energy theorem. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show potential/kinetic energy. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show conservation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show power.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 6. System of Particles and Rotational Motion

Coverage: ■ Centre of mass; torque; angular momentum; moment of inertia.

Understanding

1. What is ■ Centre of mass, and what are its main features? Use a simple example.
2. What is torque, and what are its main features? Use a school-life example.
3. What is angular momentum, and what are its main features? Use a real-world example.
4. What is moment of inertia., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Centre of mass, and what are its main features? Explain in your own words.
6. What is torque, and what are its main features? Give a step-by-step response.
7. What is angular momentum, and what are its main features? Mention one common misconception.
8. What is moment of inertia., and what are its main features? State one practical application.
9. What is ■ Centre of mass, and what are its main features? Justify your response.
10. What is torque, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to angular momentum and explain each. Use a simple example.
12. Give two examples related to moment of inertia. and explain each. Use a school-life example.
13. Give two examples related to ■ Centre of mass and explain each. Use a real-world example.
14. Give two examples related to torque and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to angular momentum and explain each. Explain in your own words.
16. Give two examples related to moment of inertia. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Centre of mass and explain each. Mention one common misconception.
18. Give two examples related to torque and explain each. State one practical application.
19. Give two examples related to angular momentum and explain each. Justify your response.
20. Give two examples related to moment of inertia. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Centre of mass related to torque? Compare the two ideas. Use a simple example.
22. How is torque related to angular momentum? Compare the two ideas. Use a school-life example.
23. How is angular momentum related to moment of inertia.? Compare the two ideas. Use a real-world example.
24. How is moment of inertia. related to ■ Centre of mass? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Centre of mass related to torque? Compare the two ideas. Explain in your own words.
26. How is torque related to angular momentum? Compare the two ideas. Give a step-by-step response.
27. How is angular momentum related to moment of inertia.? Compare the two ideas. Mention one common misconception.
28. How is moment of inertia. related to ■ Centre of mass? Compare the two ideas. State one practical application.
29. How is ■ Centre of mass related to torque? Compare the two ideas. Justify your response.
30. How is torque related to angular momentum? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply angular momentum to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply moment of inertia. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Centre of mass to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply torque to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply angular momentum to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply moment of inertia. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Centre of mass to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply torque to a realistic situation. What would you do or calculate? State one practical application.
39. Apply angular momentum to a realistic situation. What would you do or calculate? Justify your response.
40. Apply moment of inertia. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Centre of mass. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of torque. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of angular momentum. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of moment of inertia.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Centre of mass. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of torque. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of angular momentum. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of moment of inertia.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Centre of mass. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of torque. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show angular momentum. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show moment of inertia.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Centre of mass. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show torque. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show angular momentum. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show moment of inertia.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Centre of mass. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show torque. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show angular momentum. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show moment of inertia.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 7. Gravitation

Coverage: ■ Universal gravitation; potential; satellites; Kepler laws.

Understanding

1. What is ■ Universal gravitation, and what are its main features? Use a simple example.
2. What is potential, and what are its main features? Use a school-life example.
3. What is satellites, and what are its main features? Use a real-world example.
4. What is Kepler laws., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Universal gravitation, and what are its main features? Explain in your own words.
6. What is potential, and what are its main features? Give a step-by-step response.
7. What is satellites, and what are its main features? Mention one common misconception.
8. What is Kepler laws., and what are its main features? State one practical application.
9. What is ■ Universal gravitation, and what are its main features? Justify your response.
10. What is potential, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to satellites and explain each. Use a simple example.
12. Give two examples related to Kepler laws. and explain each. Use a school-life example.
13. Give two examples related to ■ Universal gravitation and explain each. Use a real-world example.
14. Give two examples related to potential and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to satellites and explain each. Explain in your own words.
16. Give two examples related to Kepler laws. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Universal gravitation and explain each. Mention one common misconception.
18. Give two examples related to potential and explain each. State one practical application.
19. Give two examples related to satellites and explain each. Justify your response.
20. Give two examples related to Kepler laws. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Universal gravitation related to potential? Compare the two ideas. Use a simple example.
22. How is potential related to satellites? Compare the two ideas. Use a school-life example.
23. How is satellites related to Kepler laws.? Compare the two ideas. Use a real-world example.
24. How is Kepler laws. related to ■ Universal gravitation? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Universal gravitation related to potential? Compare the two ideas. Explain in your own words.
26. How is potential related to satellites? Compare the two ideas. Give a step-by-step response.
27. How is satellites related to Kepler laws.? Compare the two ideas. Mention one common misconception.
28. How is Kepler laws. related to ■ Universal gravitation? Compare the two ideas. State one practical application.
29. How is ■ Universal gravitation related to potential? Compare the two ideas. Justify your response.
30. How is potential related to satellites? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply satellites to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply Kepler laws. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Universal gravitation to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply potential to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply satellites to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply Kepler laws. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Universal gravitation to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply potential to a realistic situation. What would you do or calculate? State one practical application.
39. Apply satellites to a realistic situation. What would you do or calculate? Justify your response.
40. Apply Kepler laws. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Universal gravitation. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of potential. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of satellites. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of Kepler laws.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Universal gravitation. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of potential. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of satellites. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of Kepler laws.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Universal gravitation. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of potential. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show satellites. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show Kepler laws.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Universal gravitation. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show potential. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show satellites. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show Kepler laws.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Universal gravitation. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show potential. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show satellites. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show Kepler laws.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 8. Mechanical Properties of Solids

Coverage: ■ Stress/strain; elasticity; Young's modulus.

Understanding

1. What is ■ Stress/strain, and what are its main features? Use a simple example.
2. What is elasticity, and what are its main features? Use a school-life example.
3. What is Young's modulus., and what are its main features? Use a real-world example.
4. What is ■ Stress/strain, and what are its main features? Show the idea with a diagram where appropriate.
5. What is elasticity, and what are its main features? Explain in your own words.
6. What is Young's modulus., and what are its main features? Give a step-by-step response.
7. What is ■ Stress/strain, and what are its main features? Mention one common misconception.
8. What is elasticity, and what are its main features? State one practical application.
9. What is Young's modulus., and what are its main features? Justify your response.
10. What is ■ Stress/strain, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to elasticity and explain each. Use a simple example.
12. Give two examples related to Young's modulus. and explain each. Use a school-life example.
13. Give two examples related to ■ Stress/strain and explain each. Use a real-world example.
14. Give two examples related to elasticity and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to Young's modulus. and explain each. Explain in your own words.
16. Give two examples related to ■ Stress/strain and explain each. Give a step-by-step response.
17. Give two examples related to elasticity and explain each. Mention one common misconception.
18. Give two examples related to Young's modulus. and explain each. State one practical application.
19. Give two examples related to ■ Stress/strain and explain each. Justify your response.
20. Give two examples related to elasticity and explain each. Include the relevant rule or principle.

Comparison

21. How is Young's modulus. related to ■ Stress/strain? Compare the two ideas. Use a simple example.
22. How is ■ Stress/strain related to elasticity? Compare the two ideas. Use a school-life example.
23. How is elasticity related to Young's modulus.? Compare the two ideas. Use a real-world example.
24. How is Young's modulus. related to ■ Stress/strain? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Stress/strain related to elasticity? Compare the two ideas. Explain in your own words.
26. How is elasticity related to Young's modulus.? Compare the two ideas. Give a step-by-step response.
27. How is Young's modulus. related to ■ Stress/strain? Compare the two ideas. Mention one common misconception.
28. How is ■ Stress/strain related to elasticity? Compare the two ideas. State one practical application.
29. How is elasticity related to Young's modulus.? Compare the two ideas. Justify your response.
30. How is Young's modulus. related to ■ Stress/strain? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Stress/strain to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply elasticity to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply Young's modulus. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Stress/strain to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply elasticity to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply Young's modulus. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Stress/strain to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply elasticity to a realistic situation. What would you do or calculate? State one practical application.
39. Apply Young's modulus. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Stress/strain to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of elasticity. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of Young's modulus.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Stress/strain. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of elasticity. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of Young's modulus.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Stress/strain. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of elasticity. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of Young's modulus.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Stress/strain. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of elasticity. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show Young's modulus.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Stress/strain. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show elasticity. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Young's modulus.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Stress/strain. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show elasticity. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show Young's modulus.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Stress/strain. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show elasticity. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show Young's modulus.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 9. Mechanical Properties of Fluids

Coverage: ■ Pressure; Pascal; buoyancy; Bernoulli; viscosity.

Understanding

1. What is ■ Pressure, and what are its main features? Use a simple example.
2. What is Pascal, and what are its main features? Use a school-life example.
3. What is buoyancy, and what are its main features? Use a real-world example.
4. What is Bernoulli, and what are its main features? Show the idea with a diagram where appropriate.
5. What is viscosity., and what are its main features? Explain in your own words.
6. What is ■ Pressure, and what are its main features? Give a step-by-step response.
7. What is Pascal, and what are its main features? Mention one common misconception.
8. What is buoyancy, and what are its main features? State one practical application.
9. What is Bernoulli, and what are its main features? Justify your response.
10. What is viscosity., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Pressure and explain each. Use a simple example.
12. Give two examples related to Pascal and explain each. Use a school-life example.
13. Give two examples related to buoyancy and explain each. Use a real-world example.
14. Give two examples related to Bernoulli and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to viscosity. and explain each. Explain in your own words.
16. Give two examples related to ■ Pressure and explain each. Give a step-by-step response.
17. Give two examples related to Pascal and explain each. Mention one common misconception.
18. Give two examples related to buoyancy and explain each. State one practical application.
19. Give two examples related to Bernoulli and explain each. Justify your response.
20. Give two examples related to viscosity. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Pressure related to Pascal? Compare the two ideas. Use a simple example.
22. How is Pascal related to buoyancy? Compare the two ideas. Use a school-life example.
23. How is buoyancy related to Bernoulli? Compare the two ideas. Use a real-world example.
24. How is Bernoulli related to viscosity.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is viscosity. related to ■ Pressure? Compare the two ideas. Explain in your own words.
26. How is ■ Pressure related to Pascal? Compare the two ideas. Give a step-by-step response.
27. How is Pascal related to buoyancy? Compare the two ideas. Mention one common misconception.
28. How is buoyancy related to Bernoulli? Compare the two ideas. State one practical application.
29. How is Bernoulli related to viscosity.? Compare the two ideas. Justify your response.
30. How is viscosity. related to ■ Pressure? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Pressure to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply Pascal to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply buoyancy to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply Bernoulli to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply viscosity. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Pressure to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply Pascal to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply buoyancy to a realistic situation. What would you do or calculate? State one practical application.
39. Apply Bernoulli to a realistic situation. What would you do or calculate? Justify your response.
40. Apply viscosity. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Pressure. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of Pascal. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of buoyancy. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of Bernoulli. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of viscosity.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Pressure. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of Pascal. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of buoyancy. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of Bernoulli. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of viscosity.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Pressure. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show Pascal. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show buoyancy. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Bernoulli. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show viscosity.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Pressure. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show Pascal. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show buoyancy. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show Bernoulli. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show viscosity.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 10. Thermal Properties of Matter

Coverage: ■ Heat; temperature; expansion; calorimetry; conduction.

Understanding

1. What is ■ Heat, and what are its main features? Use a simple example.
2. What is temperature, and what are its main features? Use a school-life example.
3. What is expansion, and what are its main features? Use a real-world example.
4. What is calorimetry, and what are its main features? Show the idea with a diagram where appropriate.
5. What is conduction., and what are its main features? Explain in your own words.
6. What is ■ Heat, and what are its main features? Give a step-by-step response.
7. What is temperature, and what are its main features? Mention one common misconception.
8. What is expansion, and what are its main features? State one practical application.
9. What is calorimetry, and what are its main features? Justify your response.
10. What is conduction., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Heat and explain each. Use a simple example.
12. Give two examples related to temperature and explain each. Use a school-life example.
13. Give two examples related to expansion and explain each. Use a real-world example.
14. Give two examples related to calorimetry and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to conduction. and explain each. Explain in your own words.
16. Give two examples related to ■ Heat and explain each. Give a step-by-step response.
17. Give two examples related to temperature and explain each. Mention one common misconception.
18. Give two examples related to expansion and explain each. State one practical application.
19. Give two examples related to calorimetry and explain each. Justify your response.
20. Give two examples related to conduction. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Heat related to temperature? Compare the two ideas. Use a simple example.
22. How is temperature related to expansion? Compare the two ideas. Use a school-life example.
23. How is expansion related to calorimetry? Compare the two ideas. Use a real-world example.
24. How is calorimetry related to conduction.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is conduction. related to ■ Heat? Compare the two ideas. Explain in your own words.
26. How is ■ Heat related to temperature? Compare the two ideas. Give a step-by-step response.
27. How is temperature related to expansion? Compare the two ideas. Mention one common misconception.
28. How is expansion related to calorimetry? Compare the two ideas. State one practical application.
29. How is calorimetry related to conduction.? Compare the two ideas. Justify your response.
30. How is conduction. related to ■ Heat? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Heat to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply temperature to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply expansion to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply calorimetry to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply conduction. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Heat to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply temperature to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply expansion to a realistic situation. What would you do or calculate? State one practical application.
39. Apply calorimetry to a realistic situation. What would you do or calculate? Justify your response.
40. Apply conduction. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Heat. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of temperature. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of expansion. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of calorimetry. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of conduction.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Heat. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of temperature. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of expansion. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of calorimetry. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of conduction.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Heat. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show temperature. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show expansion. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show calorimetry. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show conduction.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Heat. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show temperature. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show expansion. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show calorimetry. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show conduction.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 11.

Thermodynamics

Coverage: ■ Zeroth/first/second laws; processes; engines; entropy concepts.

Understanding

1. What is ■ Zeroth/first/second laws, and what are its main features? Use a simple example.
2. What is processes, and what are its main features? Use a school-life example.
3. What is engines, and what are its main features? Use a real-world example.
4. What is entropy concepts., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Zeroth/first/second laws, and what are its main features? Explain in your own words.
6. What is processes, and what are its main features? Give a step-by-step response.
7. What is engines, and what are its main features? Mention one common misconception.
8. What is entropy concepts., and what are its main features? State one practical application.
9. What is ■ Zeroth/first/second laws, and what are its main features? Justify your response.
10. What is processes, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to engines and explain each. Use a simple example.
12. Give two examples related to entropy concepts. and explain each. Use a school-life example.
13. Give two examples related to ■ Zeroth/first/second laws and explain each. Use a real-world example.
14. Give two examples related to processes and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to engines and explain each. Explain in your own words.
16. Give two examples related to entropy concepts. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Zeroth/first/second laws and explain each. Mention one common misconception.
18. Give two examples related to processes and explain each. State one practical application.
19. Give two examples related to engines and explain each. Justify your response.
20. Give two examples related to entropy concepts. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Zeroth/first/second laws related to processes? Compare the two ideas. Use a simple example.
22. How is processes related to engines? Compare the two ideas. Use a school-life example.
23. How is engines related to entropy concepts.? Compare the two ideas. Use a real-world example.
24. How is entropy concepts. related to ■ Zeroth/first/second laws? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Zeroth/first/second laws related to processes? Compare the two ideas. Explain in your own words.
26. How is processes related to engines? Compare the two ideas. Give a step-by-step response.
27. How is engines related to entropy concepts.? Compare the two ideas. Mention one common misconception.
28. How is entropy concepts. related to ■ Zeroth/first/second laws? Compare the two ideas. State one practical application.
29. How is ■ Zeroth/first/second laws related to processes? Compare the two ideas. Justify your response.
30. How is processes related to engines? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply engines to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply entropy concepts. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Zeroth/first/second laws to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply processes to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply engines to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply entropy concepts. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Zeroth/first/second laws to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply processes to a realistic situation. What would you do or calculate? State one practical application.
39. Apply engines to a realistic situation. What would you do or calculate? Justify your response.
40. Apply entropy concepts. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Zeroth/first/second laws. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of processes. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of engines. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of entropy concepts.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Zeroth/first/second laws. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of processes. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of engines. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of entropy concepts.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Zeroth/first/second laws. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of processes. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show engines. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show entropy concepts.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Zeroth/first/second laws. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show processes. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show engines. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show entropy concepts.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Zeroth/first/second laws. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show processes. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show engines. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show entropy concepts.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 12. Kinetic Theory

Coverage: ■ Molecular model; gas laws; degrees of freedom.

Understanding

1. What is ■ Molecular model, and what are its main features? Use a simple example.
2. What is gas laws, and what are its main features? Use a school-life example.
3. What is degrees of freedom., and what are its main features? Use a real-world example.
4. What is ■ Molecular model, and what are its main features? Show the idea with a diagram where appropriate.
5. What is gas laws, and what are its main features? Explain in your own words.
6. What is degrees of freedom., and what are its main features? Give a step-by-step response.
7. What is ■ Molecular model, and what are its main features? Mention one common misconception.
8. What is gas laws, and what are its main features? State one practical application.
9. What is degrees of freedom., and what are its main features? Justify your response.
10. What is ■ Molecular model, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to gas laws and explain each. Use a simple example.
12. Give two examples related to degrees of freedom. and explain each. Use a school-life example.
13. Give two examples related to ■ Molecular model and explain each. Use a real-world example.
14. Give two examples related to gas laws and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to degrees of freedom. and explain each. Explain in your own words.
16. Give two examples related to ■ Molecular model and explain each. Give a step-by-step response.
17. Give two examples related to gas laws and explain each. Mention one common misconception.
18. Give two examples related to degrees of freedom. and explain each. State one practical application.
19. Give two examples related to ■ Molecular model and explain each. Justify your response.
20. Give two examples related to gas laws and explain each. Include the relevant rule or principle.

Comparison

21. How is degrees of freedom. related to ■ Molecular model? Compare the two ideas. Use a simple example.
22. How is ■ Molecular model related to gas laws? Compare the two ideas. Use a school-life example.
23. How is gas laws related to degrees of freedom.? Compare the two ideas. Use a real-world example.
24. How is degrees of freedom. related to ■ Molecular model? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Molecular model related to gas laws? Compare the two ideas. Explain in your own words.
26. How is gas laws related to degrees of freedom.? Compare the two ideas. Give a step-by-step response.
27. How is degrees of freedom. related to ■ Molecular model? Compare the two ideas. Mention one common misconception.
28. How is ■ Molecular model related to gas laws? Compare the two ideas. State one practical application.
29. How is gas laws related to degrees of freedom.? Compare the two ideas. Justify your response.
30. How is degrees of freedom. related to ■ Molecular model? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Molecular model to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply gas laws to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply degrees of freedom. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Molecular model to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply gas laws to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply degrees of freedom. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Molecular model to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply gas laws to a realistic situation. What would you do or calculate? State one practical application.
39. Apply degrees of freedom. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Molecular model to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of gas laws. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of degrees of freedom.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Molecular model. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of gas laws. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of degrees of freedom.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Molecular model. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of gas laws. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of degrees of freedom.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Molecular model. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of gas laws. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show degrees of freedom.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Molecular model. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show gas laws. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show degrees of freedom.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Molecular model. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show gas laws. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show degrees of freedom.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Molecular model. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show gas laws. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show degrees of freedom.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 13. Oscillations

Coverage: ■ SHM; pendulum; energy.

Understanding

1. What is ■ SHM, and what are its main features? Use a simple example.
2. What is pendulum, and what are its main features? Use a school-life example.
3. What is energy., and what are its main features? Use a real-world example.
4. What is ■ SHM, and what are its main features? Show the idea with a diagram where appropriate.
5. What is pendulum, and what are its main features? Explain in your own words.
6. What is energy., and what are its main features? Give a step-by-step response.
7. What is ■ SHM, and what are its main features? Mention one common misconception.
8. What is pendulum, and what are its main features? State one practical application.
9. What is energy., and what are its main features? Justify your response.
10. What is ■ SHM, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to pendulum and explain each. Use a simple example.
12. Give two examples related to energy. and explain each. Use a school-life example.
13. Give two examples related to ■ SHM and explain each. Use a real-world example.
14. Give two examples related to pendulum and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to energy. and explain each. Explain in your own words.
16. Give two examples related to ■ SHM and explain each. Give a step-by-step response.
17. Give two examples related to pendulum and explain each. Mention one common misconception.
18. Give two examples related to energy. and explain each. State one practical application.
19. Give two examples related to ■ SHM and explain each. Justify your response.
20. Give two examples related to pendulum and explain each. Include the relevant rule or principle.

Comparison

21. How is energy. related to ■ SHM? Compare the two ideas. Use a simple example.
22. How is ■ SHM related to pendulum? Compare the two ideas. Use a school-life example.
23. How is pendulum related to energy.? Compare the two ideas. Use a real-world example.
24. How is energy. related to ■ SHM? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ SHM related to pendulum? Compare the two ideas. Explain in your own words.
26. How is pendulum related to energy.? Compare the two ideas. Give a step-by-step response.
27. How is energy. related to ■ SHM? Compare the two ideas. Mention one common misconception.
28. How is ■ SHM related to pendulum? Compare the two ideas. State one practical application.
29. How is pendulum related to energy.? Compare the two ideas. Justify your response.
30. How is energy. related to ■ SHM? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ SHM to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply pendulum to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply energy. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ SHM to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply pendulum to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply energy. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ SHM to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply pendulum to a realistic situation. What would you do or calculate? State one practical application.
39. Apply energy. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ SHM to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of pendulum. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of energy.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ SHM. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of pendulum. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of energy.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ SHM. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of pendulum. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of energy.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ SHM. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of pendulum. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show energy.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ SHM. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show pendulum. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show energy.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ SHM. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show pendulum. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show energy.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ SHM. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show pendulum. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show energy.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — 14. Waves

Coverage: ■ Wave motion; superposition; standing waves; Doppler effect.

Understanding

1. What is ■ Wave motion, and what are its main features? Use a simple example.
2. What is superposition, and what are its main features? Use a school-life example.
3. What is standing waves, and what are its main features? Use a real-world example.
4. What is Doppler effect., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Wave motion, and what are its main features? Explain in your own words.
6. What is superposition, and what are its main features? Give a step-by-step response.
7. What is standing waves, and what are its main features? Mention one common misconception.
8. What is Doppler effect., and what are its main features? State one practical application.
9. What is ■ Wave motion, and what are its main features? Justify your response.
10. What is superposition, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to standing waves and explain each. Use a simple example.
12. Give two examples related to Doppler effect. and explain each. Use a school-life example.
13. Give two examples related to ■ Wave motion and explain each. Use a real-world example.
14. Give two examples related to superposition and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to standing waves and explain each. Explain in your own words.
16. Give two examples related to Doppler effect. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Wave motion and explain each. Mention one common misconception.
18. Give two examples related to superposition and explain each. State one practical application.
19. Give two examples related to standing waves and explain each. Justify your response.
20. Give two examples related to Doppler effect. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Wave motion related to superposition? Compare the two ideas. Use a simple example.
22. How is superposition related to standing waves? Compare the two ideas. Use a school-life example.
23. How is standing waves related to Doppler effect.? Compare the two ideas. Use a real-world example.
24. How is Doppler effect. related to ■ Wave motion? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Wave motion related to superposition? Compare the two ideas. Explain in your own words.
26. How is superposition related to standing waves? Compare the two ideas. Give a step-by-step response.
27. How is standing waves related to Doppler effect.? Compare the two ideas. Mention one common misconception.
28. How is Doppler effect. related to ■ Wave motion? Compare the two ideas. State one practical application.
29. How is ■ Wave motion related to superposition? Compare the two ideas. Justify your response.
30. How is superposition related to standing waves? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply standing waves to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply Doppler effect. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Wave motion to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply superposition to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply standing waves to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply Doppler effect. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Wave motion to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply superposition to a realistic situation. What would you do or calculate? State one practical application.
39. Apply standing waves to a realistic situation. What would you do or calculate? Justify your response.
40. Apply Doppler effect. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Wave motion. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of superposition. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of standing waves. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of Doppler effect.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Wave motion. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of superposition. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of standing waves. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of Doppler effect.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Wave motion. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of superposition. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show standing waves. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show Doppler effect.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Wave motion. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show superposition. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show standing waves. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show Doppler effect.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Wave motion. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show superposition. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show standing waves. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show Doppler effect.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — CHEMISTRY — Core

Coverage: ■ Mole concept; atomic structure; periodicity; bonding; thermodynamics; equilibrium; redox; organic basics; hydrocarbons.

Understanding

1. What is ■ Mole concept, and what are its main features? Use a simple example.
2. What is atomic structure, and what are its main features? Use a school-life example.
3. What is periodicity, and what are its main features? Use a real-world example.
4. What is bonding, and what are its main features? Show the idea with a diagram where appropriate.
5. What is thermodynamics, and what are its main features? Explain in your own words.
6. What is equilibrium, and what are its main features? Give a step-by-step response.
7. What is redox, and what are its main features? Mention one common misconception.
8. What is organic basics, and what are its main features? State one practical application.
9. What is ■ Mole concept, and what are its main features? Justify your response.
10. What is atomic structure, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to periodicity and explain each. Use a simple example.
12. Give two examples related to bonding and explain each. Use a school-life example.
13. Give two examples related to thermodynamics and explain each. Use a real-world example.
14. Give two examples related to equilibrium and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to redox and explain each. Explain in your own words.
16. Give two examples related to organic basics and explain each. Give a step-by-step response.
17. Give two examples related to ■ Mole concept and explain each. Mention one common misconception.
18. Give two examples related to atomic structure and explain each. State one practical application.
19. Give two examples related to periodicity and explain each. Justify your response.
20. Give two examples related to bonding and explain each. Include the relevant rule or principle.

Comparison

21. How is thermodynamics related to equilibrium? Compare the two ideas. Use a simple example.
22. How is equilibrium related to redox? Compare the two ideas. Use a school-life example.
23. How is redox related to organic basics? Compare the two ideas. Use a real-world example.
24. How is organic basics related to ■ Mole concept? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Mole concept related to atomic structure? Compare the two ideas. Explain in your own words.
26. How is atomic structure related to periodicity? Compare the two ideas. Give a step-by-step response.
27. How is periodicity related to bonding? Compare the two ideas. Mention one common misconception.
28. How is bonding related to thermodynamics? Compare the two ideas. State one practical application.
29. How is thermodynamics related to equilibrium? Compare the two ideas. Justify your response.
30. How is equilibrium related to redox? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply redox to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply organic basics to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Mole concept to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply atomic structure to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply periodicity to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply bonding to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply thermodynamics to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply equilibrium to a realistic situation. What would you do or calculate? State one practical application.
39. Apply redox to a realistic situation. What would you do or calculate? Justify your response.
40. Apply organic basics to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Mole concept. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of atomic structure. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of periodicity. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of bonding. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of thermodynamics. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of equilibrium. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of redox. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of organic basics. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Mole concept. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of atomic structure. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show periodicity. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show bonding. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show thermodynamics. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show equilibrium. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show redox. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show organic basics. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mole concept. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show atomic structure. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show periodicity. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show bonding. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — BIOLOGY — Core

Coverage: ■ Living world; biological classification; plant/animal kingdoms; morphology/anatomy; cell; biomolecules; cell division;

Understanding

1. What is ■ Living world, and what are its main features? Use a simple example.
2. What is biological classification, and what are its main features? Use a school-life example.
3. What is plant/animal kingdoms, and what are its main features? Use a real-world example.
4. What is morphology/anatomy, and what are its main features? Show the idea with a diagram where appropriate.
5. What is cell, and what are its main features? Explain in your own words.
6. What is biomolecules, and what are its main features? Give a step-by-step response.
7. What is cell division, and what are its main features? Mention one common misconception.
8. What is ■ Living world, and what are its main features? State one practical application.
9. What is biological classification, and what are its main features? Justify your response.
10. What is plant/animal kingdoms, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to morphology/anatomy and explain each. Use a simple example.
12. Give two examples related to cell and explain each. Use a school-life example.
13. Give two examples related to biomolecules and explain each. Use a real-world example.
14. Give two examples related to cell division and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to ■ Living world and explain each. Explain in your own words.
16. Give two examples related to biological classification and explain each. Give a step-by-step response.
17. Give two examples related to plant/animal kingdoms and explain each. Mention one common misconception.
18. Give two examples related to morphology/anatomy and explain each. State one practical application.
19. Give two examples related to cell and explain each. Justify your response.
20. Give two examples related to biomolecules and explain each. Include the relevant rule or principle.

Comparison

21. How is cell division related to ■ Living world? Compare the two ideas. Use a simple example.
22. How is ■ Living world related to biological classification? Compare the two ideas. Use a school-life example.
23. How is biological classification related to plant/animal kingdoms? Compare the two ideas. Use a real-world example.
24. How is plant/animal kingdoms related to morphology/anatomy? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is morphology/anatomy related to cell? Compare the two ideas. Explain in your own words.
26. How is cell related to biomolecules? Compare the two ideas. Give a step-by-step response.
27. How is biomolecules related to cell division? Compare the two ideas. Mention one common misconception.
28. How is cell division related to ■ Living world? Compare the two ideas. State one practical application.
29. How is ■ Living world related to biological classification? Compare the two ideas. Justify your response.
30. How is biological classification related to plant/animal kingdoms? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply plant/animal kingdoms to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply morphology/anatomy to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply cell to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply biomolecules to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply cell division to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Living world to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply biological classification to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply plant/animal kingdoms to a realistic situation. What would you do or calculate? State one practical application.
39. Apply morphology/anatomy to a realistic situation. What would you do or calculate? Justify your response.
40. Apply cell to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

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49. A student gives an incorrect explanation of cell division. Identify a likely error and explain the correction. Justify your response.
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Higher-Order / Activity

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60. Design a short activity, investigation, problem, model, or demonstration to test or show morphology/anatomy. Include the relevant rule or principle.